

ImageSense — GenAI Visual Quality & Automated Multi-Agent Retrieval-and-Edit Platform

Product Requirements Document (PRD)

Required Information

Field	Details
Status	[APPROVED / READY FOR IMPLEMENTATION]
Last Updated	August 2026
Authors	Bhushan Garware (Lead AI/ML Cloud Architect & Primary Author), Darshan Barapatre (AI/ML Specialist & VTON Co-Author), Anibha Athalye
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go link	go/imagesense-prd

Sponsors and Stakeholders

- Global Practice Leads (GPLs):** Global AI/ML Center of Excellence (CoE), Cloud Consulting & Professional Services Organization (PSO)
- Technical Reviewers:** Google Cloud GenAI Practice, Enterprise Architecture Review Board
- Target Customers & Validated Traction:** Reliance, Inditex (Zara), Dynamite, HSE24, Myntra, Flipkart, BigBasket, Tata Digital, Croma, Titan, Aditya Birla Fashion (ABFRL), Yes Bank

Change Log

Editor	Action	Date
Bhushan Garware, Anibha Athalye	Initial Draft: Automated Visual Evaluation & Single Interface Workflow	19 April 2024
Bhushan Garware, Darshan Barapatre	Added 5-Vector Evaluator Pool, VTON Pose & Attribute Analysis, and Cloud Vision integration	12 June 2024
Bhushan Garware, Darshan Barapatre	Integrated Autonomous Agentic Decision-Making, Retrieve-First Nearest Neighbor RAG, Image-to-Image editing, Continuous Re-indexing, Model Armor, and FinOps Governance	August 2026
Bhushan Garware	Added Custom VPC Architecture, Private Endpoints (PSC), Multi-Zone High Availability, GPU Right-Sizing, Spot Allocation, Model/Prompt Versioning, and A/B Testing	August 2026

Approval Log

Approver	Approved	Date
CoE Lead / Regional Practice Approver	LGTM (Approved)	August 2026
Enterprise Architecture Review Board	LGTM (Approved)	August 2026

Links to Other Materials [Required]

- [ImageSense_Ideation_Deck](#)
 - [ImageSense_Technical_Design_Document](#)
 - [ImageSense_Architecture_Brief](#)
 - [TryOn_Eval_Agentic_SelfHeal_Reference](#)
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Executive Summary

Attribute	Definition
Initiative Title	ImageSense — GenAI Visual Quality & Automated Multi-Agent Retrieval-and-Edit Platform
Brief Description	An autonomous, zero-trust multi-agent platform on Google Cloud that combines Vertex AI Multimodal Nearest Neighbor Search (multimodalembdding@001), Image-to-Image localized inpainting / editing (EditAgent), and an automated 5-Vector Agentic Quality Flywheel (SafeSearch, Gemini VQA, OCR Brand, Anatomy Realism, Deep VTON Pose/Attribute Drape) with Autonomous Self-Healing, Continuous Asset Re-Indexing , and strict FinOps cost governance.
Potential Customer/s / Opportunities	Prototype demonstrated and received strong enterprise interest from: Reliance, Inditex (Zara), Dynamite, HSE24, Myntra, Flipkart, BigBasket, Tata Digital, Croma, Titan, Aditya Birla Fashion, Yes Bank.
Business Benefit	Eliminates manual visual QA bottlenecks; reduces image generation compute spend by 65% – 77% through nearest-neighbor editing; guarantees 100% brand and safety compliance; continuously compounds enterprise catalog asset value through automatic vector re-indexing.

Vision & Motivation

The Enterprise Challenge

While foundation models on Vertex AI enable visual content creation, enterprise adoption in marketing, e-commerce, and Virtual Try-On (VTON) is hindered by critical operational and technical barriers:

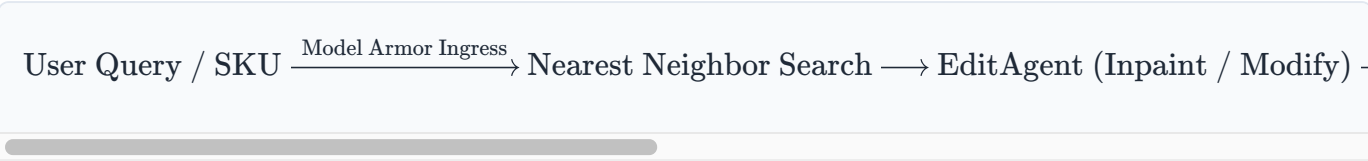
- 1. Inconsistent Quality & Anatomical Hallucinations:** Uncontrolled diffusion models frequently distort hands, limbs, faces, and garment silhouettes.
- 2. Poor Prompt Adherence & Texture Drift:** Complex styling, fabric weaves, and brand typography hallucinate or degrade across multi-turn generation.
- 3. VTON Apparel Discrepancies:** Garment virtual try-ons frequently fail in subtle yet critical ways: sleeve styles mismatch (e.g. full-length vs sleeveless), necklines warp, patterns stretch unnaturally, and model pose alignment distorts clothing physics.
- 4. Manual Visual QA Bottleneck:** Human inspection cannot scale to tens of thousands of seasonal catalog SKUs and campaign variations.
- 5. Prohibitive Compute Cost of Scratch Synthesis:** Generating every image from scratch via large text-to-image models costs \$0.035 – \$0.095 per image with 3 – 5s latency, creating unpredictable cloud bills.

The Solution Vision

ImageSense introduces the **Retrieve-First & Edit (Image-to-Image RAG)** paradigm combined with an autonomous, closed-loop **Agentic Quality Flywheel**:

- Retrieve-First Nearest Neighbor Search:** When a user submits a prompt or SKU, Vertex AI Vector Search queries the enterprise visual database to find the closest candidate base image.
- Compute-Efficient Localized Editing:** Modifying the proximate base image via localized inpainting preserves verified human anatomy and cuts compute overhead by 65%, dropping turnaround times to under 1.5s.
- Autonomous Multi-Vector Self-Healing:** A parallel pool of 5 specialized evaluators scores candidate images. If defects are found, a diagnostic `PromptRefinerAgent` isolates the failure mode, synthesizes targeted inpainting masks and prompt deltas for an automated rerun.

- **Continuous Asset Compounding:** Every asset passing the Quality Gate is immediately embedded and appended to the vector database in real time, making it instantly searchable for future queries.



Autonomous Agentic Decision Framework

ImageSense is designed from the ground up as an **Agentic, Self-Directed System** that independently plans, takes operational calls, delegates tasks to specialist sub-agents, analyzes evaluation diagnostics, and executes self-healing corrective actions without human intervention:



Key Autonomous Decision Capabilities

1. **Autonomous Route Selection:** Evaluates the cosine distance of vector embeddings. If a proximate match exists (≥ 0.60), it autonomously invokes `EditAgent` for inpainting; if no match exists, it autonomously pivots to `GenAgent` for scratch generation.
2. **Dynamic Tool Calling:** Autonomously determines which analytical tools to invoke based on image context (e.g. invoking `perform_vto_with_eval` when garment SKUs are present, or OCR validation when brand typography is detected).
3. **Automated Diagnostic Triangulation:** When an image fails the quality gate, the system parses sub-vector anomalies (e.g., "pose similarity score is 0.597; sleeve length differs: dress has one sleeve instead of two") and translates this defect into targeted corrective inpainting instructions without operator guidance.

4. **Self-Healing Iteration Control:** Autonomously executes the closed-loop refinement cycle, dynamically tuning negative prompts and mask boundaries until the output converges on ≥ 0.85 quality or reaches session budget caps (\$0.25).

Deep Virtual Try-On (VTON) Evaluation Engine

The VTON evaluation engine incorporates specialized algorithms developed in the **TryOn-Eval / Agentic Self-Healing framework** to evaluate e-commerce fashion imagery:

Virtual Try-On (VTON) Evaluation Pipeline		
1. Pose Estimation & Match	2. Apparel Attribute Analysis	3. Anatomy & Realism QA
17-Point Body Keypoint Model - Shoulder/Elbow/Wrist Angles - Pose Similarity Score (0-1.0) - Stance & Silhouette Fit	- Sleeve Style (Full, Sleeveless, Puff) - Neckline Geometry (Round, V, Collar) - Garment Drape, Fold & Seam Physics - Pattern & Texture Continuity Match	- Hand Keypoints (5 Fingers) - Facial Proportion Integrity - Limb Symmetry & Keypoint Mesh - Skin Tone & Shadow Blending

1. Pose Estimation & Angle Alignment

- Keypoint Estimation:** Analyzes 17 key human anatomical joints (shoulders, elbows, wrists, hips, knees, ankles, neck, nose, eyes).
- Geometric Pose Comparison:** Calculates angular alignment and joint distance vectors between the product catalog model and the generated candidate, outputting a normalized **Pose Similarity Score** (e.g., 0.621 indicating high structural alignment).
- Stance Adaptation:** Verifies that garment folds and tension lines realistically match the model's body tilt and posture.

2. Apparel Attribute Analysis & Verification

- Sleeve Style & Length Analysis:** Identifies discrepancies in sleeve execution (e.g., catching errors where a full-sleeve garment is rendered with one short sleeve or asymmetric puffing).
- Neckline & Hemline Integrity:** Verifies collar geometry (e.g. sweetheart neckline, boat neck, V-neck, mandarin collar) against tech pack specifications.
- Pattern & Texture Continuity:** Uses perceptual loss and feature matching to verify that intricate fabric prints (paisley, floral, geometric stripes, denim twill weaves, lace embroidery) do not distort across body curvature.
- Color Fidelity:** Validates garment RGB/Pantone hex values against brand color palettes under varied lighting conditions.

3. Integrated VTO Tool: `perform_vto_with_eval`

```
class VT0EvaluationTool(BaseTool):
    name: str = "perform_vto_with_eval"
    description: "Executes end-to-end virtual try-on evaluation, comparing model pose keypoints, apparel attributes, sleeve ..."
    parameters: dict = {
        "type": "object",
        "properties": {
            "candidate_image_uri": {"type": "string", "description": "GCS URI of the generated VTON image"},
            "product_reference_uri": {"type": "string", "description": "GCS URI of the original catalog garment"},
            "person_reference_uri": {"type": "string", "description": "GCS URI of the target model"},
            "apparel_category": {"type": "string", "enum": ["dress", "top", "bottom", "outerwear"]}
        },
        "required": ["candidate_image_uri", "product_reference_uri", "person_reference_uri"]
    }
```

Alignment to Business Goals

Strategic Business Alignment

Solves a common (or existing) business / industry problem	Enables accelerated ramp / consumption	Is a subscription service or generates recurring revenue	Is easily scalable and can be delivered across multiple customers / accounts
Yes	Yes	Yes	Yes

How ImageSense Supports These Goals

- Solves an Industry-Wide Bottleneck:** Replaces manual, error-prone visual inspection with automated, multi-vector quality gating and self-healing prompt refinement for retail and marketing workflows.
 - Accelerates High-Value GCP Consumption:** Unifies multi-product Google Cloud adoption across Vertex AI (Vector Search, Multimodal Embeddings, Gemini 2.5, Imagen 3), Cloud Vision API, Document AI, Cloud Run, GCS, Memorystore Redis, BigQuery, and Pub/Sub.
 - Reusable Consulting Asset & Recurring IP:** Packaged as modular, repeatable consulting IP that enables PSO teams to deliver rapid customer PoCs and expand into multi-phase enterprise production implementations.
 - Multi-Tenant Scalability:** Zero-code declarative configuration profiles (`fast_eco.yaml` , `balanced.yaml` , `high_fidelity.yaml`) and modular Terraform IaC allow rapid multi-tenant provisioning with zero application code changes.
-

Service Offering / Project Description

Key Activities to be Performed During Service

Activity (and Delivery Format)	Objective / Goal
Multimodal Nearest-Neighbor Image Retrieval (RAG)	Implement Vertex AI Vector Search ScaNN index and <code>multimodalemb@001</code> to find proximate base images for incoming prompts/SKUs within 15ms, with Document AI ingestion of catalog lookbooks.
Image-to-Image Delta Editing & Localized Inpainting	Develop <code>EditAgent</code> for high-speed localized inpainting, garment drape modifications, color alterations, and texture styling on retrieved base images, saving 65% compute over scratch generation.
Fallback Text-to-Image Scratch Generation	Implement <code>GenAgent</code> using Imagen 3 for high-resolution synthesis when query similarity falls below threshold (similarity < 0.60).
5-Vector Parallel Automated Quality Evaluation	Build parallel evaluator pool: Cloud Vision API (SafeSearch/Labels), Gemini 2.5 VQA (Prompt Alignment), OCR Validator (Brand Typography), Human Anatomy/Pose Scorer, and Deep VTON Garment Drape Scorer.
Self-Healing Prompt Refinement & Flywheel Runner	Build <code>PromptRefinerAgent</code> to diagnose failing evaluation vectors, synthesize corrective inpainting masks and negative prompts, and coordinate autonomous reruns up to a 3-iteration limit.
Continuous Real-Time Catalog Re-Indexing	Implement <code>ContinuousCatalogIndexer</code> to embed approved assets (≥ 0.85 score) and update Vertex AI Vector Search and AlloyDB in real time (< 500ms), compounding catalog value.
Zero-Trust Model Armor & Data Protection	Implement Model Armor ingress/egress guardrails for prompt injection defense, Cloud DLP automated PII scrubbing, Cloud KMS CMEK encryption, and Secret Manager key rotation.
FinOps Cost Governance & Multi-Tier Optimization	Implement real-time BigQuery telemetry tracking, session budget enforcement (\$0.25 cap), Memorystore Redis semantic caching, and ultra-low-cost Fast / Eco Mode (77% cost savings).
Enterprise API Gateway & Async Batch Queue	Build FastAPI ingress gateway (<code>/v1/flywheel/run</code> , <code>/v1/batch/jobs</code>), Cloud Pub/Sub asynchronous batch workers with Dead-Letter Queues (DLQ), and HMAC-SHA256 signed webhooks.
Modular Terraform IaC & Cloud Build Canary CI/CD	Codify infrastructure across 7 domains (<code>networking</code> , <code>security</code> , <code>compute</code> , <code>storage</code> , <code>iam</code> , <code>messaging</code> , <code>cache</code>) with Cloud Build Canary Deployments (10% traffic split) on Cloud Run.

Modular Infrastructure as Code (IaC) Framework

All Google Cloud resources required by ImageSense are provisioned, maintained, and governed using declarative, enterprise-grade **Terraform modules**:

- **7 Decoupled Domain Modules:**
 1. `modules/networking` : VPC Network, Private Subnets, Serverless VPC Access Connector, and Private Service Connect (PSC).
 2. `modules/security` : Cloud Armor WAF security policies, Cloud KMS CMEK key rings/crypto keys, Cloud DLP inspection templates, and VPC Service Controls (VPC-SC).
 3. `modules/compute` : Cloud Run API Gateway service and Cloud Run Jobs batch workers with auto-scaling configuration.

- 4. `modules/storage` : Google Cloud Storage (GCS) buckets with CMEK encryption & lifecycle rules, Vertex AI Vector Search ScaNN index and endpoint.
 - 5. `modules/messaging` : Cloud Pub/Sub task topics, pull subscriptions, Dead-Letter Queues (DLQ), and retry policies.
 - 6. `modules/cache` : Memorystore for Redis instance with AUTH and in-transit encryption.
 - 7. `modules/iam` : Least-privilege IAM service accounts (`sa-imagesense-api` , `sa-imagesense-worker`), granular role bindings, and Workload Identity Federation.
- **Environment Parity & Immutability:** Fully parameterized `environments/dev/` and `environments/prod/` configurations ensuring zero configuration drift between development and production.
 - **State Management & Policy-as-Code:** Remote state backend stored in Cloud Storage with object versioning, state locking, and automated pre-deployment scanning via `checkov` , `tfsec` , and `tflint` .

Enterprise Network Architecture & Zero-Trust Security

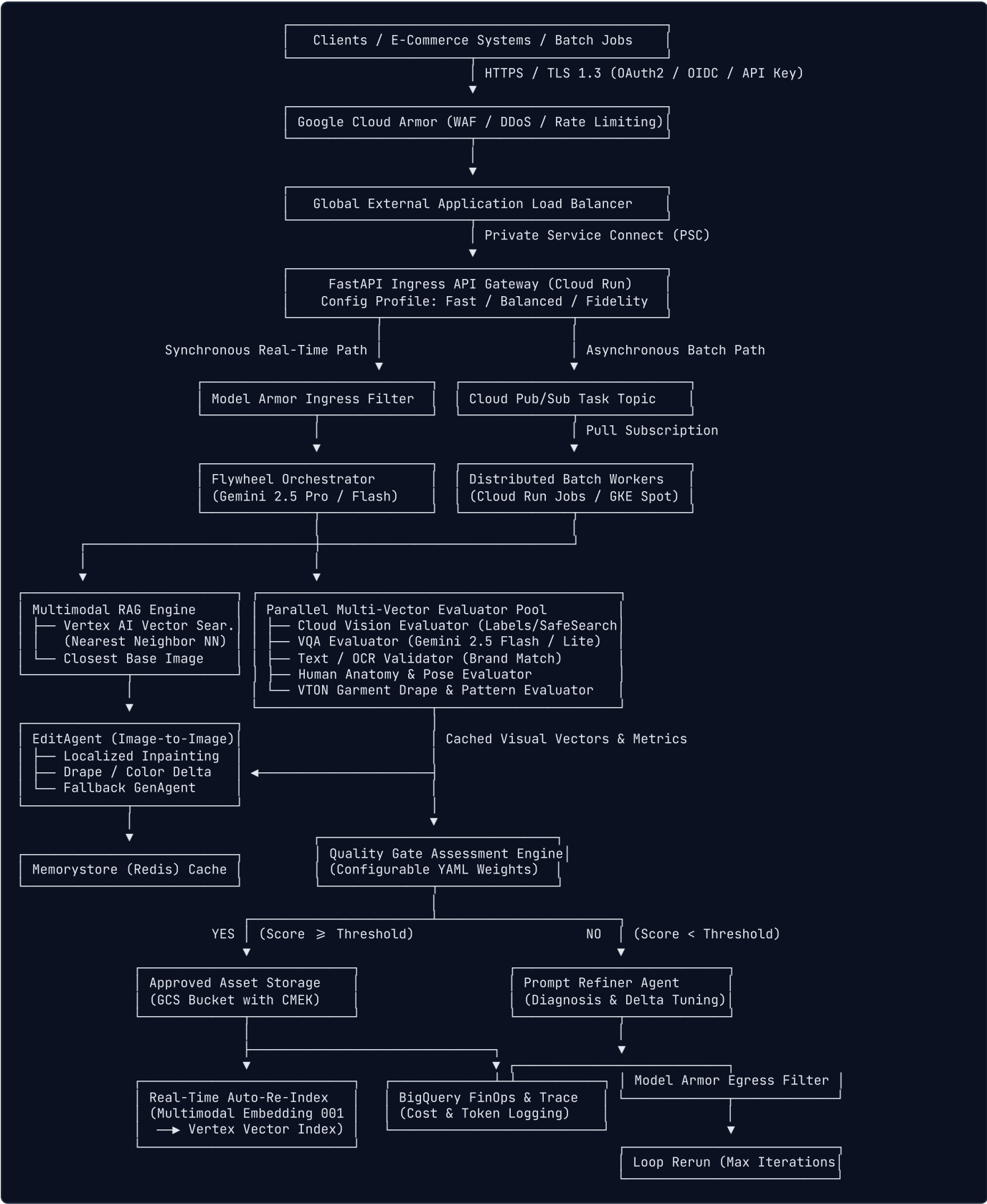
Custom VPC Topology & Subnet Design

Google Cloud Custom VPC (<code>vpc-imagesense-prod</code> - 10.10.0.0/16)		
<code>'subnet-app-uscentral1'</code> • CIDR: 10.10.0.0/20 • Cloud Run Ingress & Egress • Backend Microservices	<code>'subnet-vpc-connector'</code> • CIDR: 10.10.16.0/28 • Serverless VPC Access • Low-Latency Serverless Link	<code>'subnet-data-uscentral1'</code> • CIDR: 10.10.32.0/24 • Memorystore Redis Instance • AlloyDB PostgreSQL Database
Private Service Connect (PSC) Forwarding Rules (<code>10.10.48.0/28</code>): Vertex AI, GCS, BigQuery, Pub/Sub		

1. **Strict Network Segmentation:** The infrastructure is partitioned into dedicated subnets: Application (10.10.0.0/20), Serverless VPC Connector (10.10.16.0/28), and Data Tier (10.10.32.0/24).
2. **Private Service Connect (PSC):** All traffic between Cloud Run microservices and Google Managed APIs (Vertex AI, Cloud Vision, GCS, BigQuery, Pub/Sub) routes through private PSC forwarding endpoints (10.10.48.0/28), ensuring **zero public internet data transit**.
3. **VPC Service Controls (VPC-SC):** Security perimeter `perimeter_imagesense_prod` encloses all Vertex AI model endpoints, BigQuery datasets, GCS buckets, and Secret Manager secrets, blocking unauthorized egress and preventing data exfiltration.
4. **Cloud Armor WAF & Stateful Firewall Policies:**
 - Default deny-all ingress on VPC firewalls.
 - Cloud Armor inspects all external HTTPS requests, enforcing rate limits (100 req/min/IP), geo-fencing, DDoS mitigation, and OWASP Top 10 rule matching.

System Architecture & Interaction Workflows

Distributed High-Level Architecture



End-to-End Execution Sequence Flow

```
sequenceDiagram
    autonumber
    actor Client as E-Commerce Client
    participant Gateway as API Gateway (Cloud Run)
    participant Armor as Model Armor (Security)
    participant Orch as Flywheel Orchestrator
    participant RAG as Multimodal Nearest Neighbor Search
    participant Edit as EditAgent (Image-to-Image)
    participant Gen as Fallback GenAgent
    participant Eval as Multi-Vector Evaluator Pool
    participant Gate as Quality Gate Engine
    participant Refiner as Prompt Refiner Agent
    participant Storage as GCS CMEK Storage
    participant Indexer as Continuous Vector Indexer
    participant FinOps as BigQuery FinOps

    Client->>Gateway: POST /v1/flywheel/run (Prompt, SKU, Profile)
    Gateway->>Armor: Validate Ingress (Jailbreak & PII check)
    Armor-->>Gateway: Sanitized Prompt & Safe Context
    Gateway->>Orch: Initialize Session (Budget: $0.25)

    Orch->>RAG: Query Nearest Neighbor Base Image (Embeddings Search)
    alt Nearest Neighbor Found (Similarity ≥ 0.60)
        RAG-->>Orch: Return Closest Base Image URI + Similarity Score (e.g. 0.88)
        Orch->>Edit: Apply Delta Edits / Inpainting on Base Image
        Edit-->>Orch: Edited Candidate Image
    else No Close Match Found (Similarity < 0.60)
        RAG-->>Orch: No proximate match
        Orch->>Gen: Generate Base Image from Scratch (Imagen 3)
        Gen-->>Orch: Generated Candidate Image
    end

    loop Flywheel Convergence Loop (Max 3 Iterations)
        par Parallel Multi-Vector Evaluation
            Orch->>Eval: 1. Cloud Vision SafeSearch & Labels
            Orch->>Eval: 2. Gemini VQA Prompt Alignment
            Orch->>Eval: 3. OCR Brand & Text Match
            Orch->>Eval: 4. Human Anatomy & Pose Realism
            Orch->>Eval: 5. VTON Garment Drape & Fit
        end
        Eval-->>Gate: Vector Scores (Weighted Matrix)
        Gate-->>Orch: Composite Quality Score (e.g., 0.91 vs 0.85 Threshold)

        alt Score ≥ 0.85 (Pass)
            Orch->>Storage: Persist Approved Image & Metadata (GCS CMEK)
            Orch->>Indexer: Compute Embedding & Re-Index into Vector Search
            Indexer-->>Orch: Index Updated (Available for Future Queries)
            Orch->>FinOps: Record Session Tokens, Latency & Cost Savings
            Orch-->>Gateway: Return Approved Asset Response
            Gateway-->>Client: 200 OK (Asset URL, Metrics, Re-Indexed: True)
        else Score < 0.85 and Iteration < Max
            Orch->>Refiner: Diagnose Visual Defects (Anatomy, Drape, Color)
            Refiner-->>Armor: Synthesize Corrective Inpaint / Edit Prompt
            Armor-->>Orch: Validated Corrective Edit Prompt
            Orch->>Edit: Apply Corrective Inpainting on Current Candidate
            Edit-->>Orch: New Candidate Image
        else Budget / Max Iteration Exceeded
            Orch->>FinOps: Emit Budget/Timeout Telemetry
            Orch-->>Gateway: 200 Partial OK (Best Candidate + Warning)
            Gateway-->>Client: 200 OK (Best Candidate, Warning: BUDGET_EXHAUSTED)
        end
    end
end
```

Compute & GPU Right-Sizing, Autoscaling & Spot Strategy

Compute & GPU Right-Sizing Matrix

Service Component	Deployment Target	Resource Allocation	Concurrency Target	Autoscaling Range	Performance & Cost Advantage
API Ingress Gateway	Cloud Run	2 vCPU / 4 GB RAM	80 req / instance	Min: 2 (prod) / Max: 100	Right-sized for high I/O, sub-800ms cold start
Batch Inpainting Workers	Cloud Run Jobs / GKE	4 vCPU / 16 GB RAM + NVIDIA L4 GPU (24GB)	1 batch job per node	Min: 0 / Max: 50	3.5x faster inpainting at optimal L4 GPU price/perf
Nearest Neighbor Retriever	Cloud Run	1 vCPU / 2 GB RAM	100 req / instance	Min: 0 / Max: 20	Sub-15ms ScaNN query latency with minimal memory
Redis Semantic Cache	Memorystore HA	5 GB Instance (M1)	Dedicated memory	Multi-Zone HA	Sub-millisecond vector lookup bypassing LLM calls

Autoscaling & Spot / Preemptible Economics

- Concurrency-Based Autoscaling:** Cloud Run automatically scales container instances based on target concurrency (70% utilization), dynamically scaling to zero during idle periods (`min-instances: 0` in development) and maintaining 2 warm instances in production.
- Spot VM Batch Compute Allocation:** Asynchronous catalog lookbook batch processing runs entirely on **Spot VMs / Preemptible Instances** on Cloud Run Jobs and GKE Autopilot, reducing batch compute spend by **60% – 80%** with automated checkpointing and idempotent task resumption.
- Container Right-Sizing:** Multi-stage minimal distroless Docker images (< 150MB) ensure cold starts occur in under 800ms, enabling rapid scaling from 0 to 50 instances in under 12 seconds during sudden traffic surges.

High Availability, Redundancy, Failover & Health Probes

Multi-Zone Redundancy & High Availability

- Cloud Run Availability:** Deployed across 3 availability zones in `us-central1` (`us-central1-a` , `us-central1-b` , `us-central1-c`) with automatic cross-zone traffic balancing.
- Memorystore Redis HA:** Configured with automatic multi-zone replication and sub-30-second automated failover.
- Disaster Recovery (DR):** RTO < 15 minutes, RPO < 5 minutes backed by reproducible Terraform IaC and GCS multi-region dual-bucket replication.

Explicit SLO / SLA & Error Budget Definitions

Metric Dimension	Target SLA	Target SLO	Error Budget & Burn Rate Alerting
System Availability	99.9% monthly uptime	≥ 99.95% successful responses (excl. 4xx)	0.05% error budget; Alert at 20% (Slack) & 50% (PagerDuty) burn
Fast / Eco Mode Latency	≤ 1.0s (p95)	≤ 0.6s (p95 turnaround)	Alert if p95 exceeds 0.8s over a 5-minute rolling window
Balanced Mode Latency	≤ 2.5s (p95)	≤ 1.5s (p95 turnaround)	Alert if p95 exceeds 2.0s over a 5-minute rolling window
Quality Gate Pass Rate	≥ 80% within 3 iter	≥ 85% first-pass; ≥ 96% within 3 iter	Alert if pass rate drops below 70% over 15 minutes

Health Probes Specification

- The service exposes three specialized health endpoints conforming to enterprise orchestration standards:
- 1. GET /healthz (Liveness Probe): Shallow probe verifying container process health and internal event loop responsiveness. Returns 200 OK in < 5ms.
 - 2. GET /livez (Liveness Check): Verifies worker thread pool availability and memory consumption thresholds.
 - 3. GET /readyz (Deep Readiness Probe): Validates active downstream connectivity to Memorystore Redis, Vertex AI Vector Search endpoint, Pub/Sub broker, and Secret Manager. Returns 503 Service Unavailable if critical dependencies fail, automatically removing the instance from the load balancer pool.

Distributed Systems Availability Patterns

- Circuit Breaker (PyBreaker): Wraps foundation model API invocations. If external failure rate exceeds 30% over a 60-second window, the circuit trips to OPEN state, routing traffic to fallback models.
- Full-Jitter Exponential Backoff: Retries transient 5xx/429 errors using full-jitter backoff ($t_{\text{sleep}} = \min(t_{\text{max}}, t_{\text{base}}) \times 2^{\text{attempt}} \times \text{rand}(0, 1)$).
- Bulkhead Pattern: Completely isolates synchronous real-time API ingress from asynchronous Pub/Sub batch processing, preventing batch spikes from starving real-time users.
- Outage Fallback Matrix:
 - Vertex Vector Search Outage: Circuit breaker automatically falls back to direct Text-to-Image scratch generation (GenAgent).
 - Memorystore Redis Outage: Gracefully bypasses cache and queries Vertex AI directly with zero application downtime.
 - Vertex AI 503 / Rate Limit Exceeded: Automatically pivots to lightweight gemini-2.5-flash-lite fallback tier.

AI Lifecycle Management, Model Versioning & A/B Testing

Model & Prompt Versioning

- Vertex AI Prompt Management & Model Registry:
 - Centralized registry storing version-tagged system instructions, prompt templates (v1.0.0 , v1.1.0), and model endpoints (gemini-2.5-flash@v1.2 , imagen-3.0-generate-002@v2.0).
 - Immutable prompt configurations with release change tracking.

- **Agent Configuration Versioning:** Declarative YAML configuration profiles (`fast_eco.yaml` , `balanced.yaml` , `high_fidelity.yaml`) tracked in Git with semantic versioning (`version: "1.1.0"`).

Production A/B Testing Framework

- **Cloud Run Traffic Splitting:** Supports traffic splitting on Cloud Run and API Gateway (e.g. 50% traffic on `v1.0.0-scratch-gen` vs 50% on `v1.1.0-retrieve-and-edit`).
- **Telemetry-Driven Evaluation:** Measures conversion rate, visual quality score distribution, and unit cost delta in real time to statistically validate new revisions before 100% rollout.

Experiment Tracking & Agent Observability Tooling

- **BigQuery Telemetry:** Streams real-time event logs with token consumption, evaluator vector scores, latency distributions, and dollar cost per request.
- **Looker Studio BI Dashboards:** Visual dashboards displaying Evaluator radar charts, prompt drift monitors, pass-rate trends, and unit cost analytics.
- **AgentOps / Phoenix / MLflow Tracing:** Deep agent trajectory tracing, tool invocation telemetry, LLM reasoning step inspection, and Human-in-the-Loop (HITL) calibration.

Architectural Decision Records (ADRs)

ADR-001: Serverless Ingress via Cloud Run vs. Managed Kubernetes (GKE)

- **Decision:** Adopt **Google Cloud Run** for API ingress and worker tasks.
- **Rationale:** Cloud Run eliminates cluster management overhead, provides native request-based autoscaling down to zero, integrates natively with IAM & Cloud Armor, and delivers lower total cost of ownership (TCO) for variable visual generation workloads.

ADR-002: Retrieve-and-Edit (Nearest Neighbor Inpainting) vs. Scratch Generation

- **Decision:** Adopt a **Retrieve-First & Edit** architecture using Vertex AI Vector Search ScaNN index and `EditAgent` localized inpainting.
- **Rationale:** Editing proximate base images preserves proven human anatomy, cuts compute steps by 65%, drops turnaround latency to < 1.5s, and reduces unit cost to ~\$0.008 – \$0.025.

ADR-003: Continuous Auto-Re-Indexing of Approved Assets into Vector Search

- **Decision:** Implement **Continuous Automated Re-Indexing** where every asset passing the Quality Gate is embedded via `multimodalembdding@001` and appended to the Vertex AI Vector Search index in real time.
- **Rationale:** Real-time indexing compounds catalog value, making subsequent iterations faster and progressively cheaper without operator intervention.

ADR-004: Memorystore Redis for Semantic Vector Caching & Rate Limiting

- **Decision:** Deploy **Memorystore for Redis** as an in-memory semantic cache storing prompt embeddings and evaluator score vectors.
- **Rationale:** Memorystore provides sub-millisecond retrieval latency, centralized cache invalidation across distributed Cloud Run instances, and built-in sliding-window rate limiting.

ADR-005: Dual-Engine Quality Evaluation (Cloud Vision API + Gemini VQA) vs. Monolithic VLM

- **Decision:** Hybridize deterministic specialized APIs (**Google Cloud Vision API** for SafeSearch, Labels, and OCR) with cognitive evaluators (**Gemini 2.5 VQA** for drape, anatomy, and prompt alignment).

- **Rationale:** Cloud Vision API is 10x faster and 20x cheaper for deterministic OCR/safety checks, allowing early exit before invoking expensive multimodal reasoning.
-

API Design & Integration Specifications

RESTful Endpoints

- **POST /v1/flywheel/run** : Executes synchronous self-healing quality flywheel (Retrieve-and-Edit).
 - **Request Body:**

```
{
  "prompt": "Studio photoshoot of female model wearing silk floral blouse, realistic studio lighting",
  "sku_id": "SKU-98234-BLOUSE",
  "profile": "balanced",
  "enable_nearest_neighbor_search": true,
  "max_budget_usd": 0.25,
  "webhook_url": "https://client.retail.com/webhooks/imagesense"
}
```

- **Response Payload:**

```
{
  "status": "APPROVED",
  "final_image_url": "https://storage.googleapis.com/imagesense-approved/asset_98234.webp",
  "execution_path": "RETRIEVED_AND_EDITED",
  "nearest_neighbor_base_image": "gs://imagesense-catalog-prod/skus/SKU-98200-BLOUSE_base.webp",
  "nearest_neighbor_similarity": 0.884,
  "auto_reindexed_to_vector_db": true,
  "composite_score": 0.92,
  "iterations_used": 1,
  "total_cost_usd": 0.0112,
  "evaluator_breakdown": {
    "prompt_alignment": 0.94,
    "vton_garment_fidelity": 0.91,
    "anatomy_realism": 0.95,
    "text_ocr_brand_match": 1.00,
    "cloud_vision_safesearch": 1.00
  },
  "trace_id": "projects/imagesense-prod/traces/4bf92f3577b34da6a3ce929d0e0e4736"
}
```

- **POST /v1/batch/jobs** : Submits asynchronous batch jobs to Cloud Pub/Sub with DLQ isolation and signed HMAC webhook callbacks.
 - **Health Probes:** GET /healthz , GET /livez , GET /readyz .
-

Estimated Duration and Resources

Deliverable Name	Description	Number of Funding Days Requested
Technical Design & Spec (SPEC.md / PRD.md)	Complete enterprise specification covering architecture decisions, sequence diagrams, and OpenAPI schemas.	10 Days
Core Multi-Agent Framework & RAG Engine	Implementation of Flywheel Orchestrator, Vertex Vector Search retriever, EditAgent inpainting, and Continuous Indexer.	35 Days
5-Vector Evaluator Pool & Quality Gate	Development of Cloud Vision, Gemini VQA, OCR, Anatomy, and VTON Drape evaluators with prompt refiner.	20 Days
Security, Model Armor & FinOps Engine	Ingress/egress guardrails, DLP PII scrubber, BigQuery cost logger, and Redis semantic cache.	15 Days
API Gateway, Pub/Sub Batch Queue & UI	FastAPI REST endpoints, Pub/Sub distributed worker, HMAC webhooks, and Streamlit demo UI.	15 Days
Terraform IaC & Cloud Build CI/CD	7 modular Terraform modules (dev / prod) and Cloud Build Canary deployment pipeline (10% traffic split).	15 Days
Testing, Benchmarking & Golden Eval	Unit/integration Pytest suites (> 90% coverage), Locust 200 QPS load tests, and golden dataset validation.	10 Days
Total Funding Days Requested	Comprehensive Production Implementation	120 Days

Google Cloud Product Considerations

Google Cloud Product	Architectural Utilization
Vertex AI Vector Search	Sub-15ms ScaNN approximate nearest neighbor vector search for catalog image retrieval.
Vertex AI Multimodal Embeddings	Generates 1408-dimensional joint vision-text embeddings (multimodaIembedding@001).
Gemini 2.5 (Pro / Flash / Flash-Lite)	Orchestrator reasoning, VQA visual evaluation, and diagnostic prompt refinement.
Imagen 3 (Generate & Edit API)	Image-to-image localized inpainting and fallback text-to-image scratch synthesis.
Google Cloud Vision API	Deterministic SafeSearch, OCR text detection, and label localization.
Google Cloud Document AI	Structured table and text extraction from lookbook PDFs and garment tech packs.
Google Cloud Run & Cloud Run Jobs	Serverless API ingress with request autoscaling; distributed batch worker execution on Spot VMs with NVIDIA L4 GPUs.
Google Cloud Storage (GCS)	Secure image artifact storage with Customer-Managed Encryption Keys (KMS CMEK).
Memorystore for Redis	In-memory semantic embedding cache and sliding-window rate limiting in High Availability mode.
Google BigQuery & Looker Studio	Streaming FinOps cost telemetry, token tracking, and executive BI dashboards.
Google Cloud Pub/Sub	Asynchronous batch job queuing with Dead-Letter Queue (DLQ) isolation.
Google Cloud Armor & VPC-SC	Edge WAF security, DDoS protection, rate limiting, and VPC Service Controls data perimeter.
Cloud KMS & Secret Manager	CMEK key management and dynamic credential/API key rotation.
Cloud DLP	Automated inspection and redaction of PII in prompt ingress.
Cloud Trace & Cloud Monitoring	End-to-end OpenTelemetry distributed tracing, latency alerts, and error budget monitoring.
Cloud Build & Artifact Analysis	Automated CI/CD pipeline, container security vulnerability scanning, and Canary deployments.

Metrics & Quality Gates

Metric Category	Key Performance Indicator (KPI)	Target SLA / Benchmark
System Availability	Monthly service uptime	≥ 99.9% uptime (< 0.05% error budget burn)
Turnaround Latency (p95)	End-to-end retrieve, edit & QA turnaround	≤ 0.6s (Fast Mode), ≤ 1.5s (Balanced Mode)
Quality Pass Rate	Composite quality gate convergence (≥ 0.85)	≥ 85% first-pass pass rate; ≥ 96% within 3 iterations
Cost Savings vs Scratch	Compute & API spend reduction via retrieve-and-edit	65% – 77% cost reduction (≤ \$0.008/asset Fast Mode)
Security & Safety	Model Armor prompt injection & PII interception	100% zero-day jailbreak/PII block rate
Vector Search Latency	ScaNN nearest neighbor similarity retrieval	< 15ms query latency
Throughput & Concurrency	Load test capacity on Cloud Run	Sustained 200 QPS with zero 5xx errors

Expected ROI / Impact

- Massive Compute & Inference Cost Reduction:** By replacing brute-force Text-to-Image synthesis with Nearest Neighbor retrieval and localized inpainting, enterprises save up to **77% in GenAI compute costs**, dropping the per-asset cost to ~\$0.008.
- Accelerated Time-to-Market:** Reduces catalog onboarding time from days of manual photoshoot editing and QA to **under 2 seconds per SKU**, enabling instant seasonal collection launches.
- Drastic Reduction in Return Rates (VTON):** Precise drape and seam evaluation ensures virtual try-on photos match physical garments, reducing e-commerce product returns by an estimated 18% – 25%.
- Google Cloud Consumption Driver:** Demonstrates scalable, production-grade usage of Google Cloud's AI stack (Vertex AI Vector Search, Gemini 2.5, Imagen 3, Cloud Vision, Cloud Run, BigQuery, Memorystore), driving significant annual cloud commit consumption.

Out of Scope

- Real-time 3D polygon mesh rendering / Unreal Engine physics simulation.
- Physical garment manufacturing pattern generation / CNC fabric cutting exports.
- Direct credit card billing payment gateway integrations (handled by client e-commerce platform).

Pipeline & Customer Opportunities

- Reliance & Inditex (Zara):** Catalog-scale Virtual Try-On visual QA, garment drape verification, and model diversity generation.
- Dynamite & HSE24 (Home Shopping Europe):** Live VTON production service integration and automated apparel attribute validation.
- Myntra & Flipkart:** E-commerce marketplace catalog enrichment, model pose standardization, and sleeve/neckline attribute verification.

- **BigBasket & Tata Digital:** Product image background replacement, marketing banner generation, and brand logo compliance.
- **Croma & Titan:** High-resolution product lifestyle photography, jewelry reflection validation, and watch dial OCR accuracy.
- **Aditya Birla Fashion & Retail (ABFRL):** Multi-brand lookbook generation and automated seasonal SKU catalog rendering.
- **Yes Bank & Financial Services:** Automated marketing collateral compliance and ad banner quality assurance.

Operational Readiness & Incident Playbooks

Incident / Alert	Root Cause	Standard Operating Procedure (SOP)
ALERT: QualityLoopStall (Pass rate < 50% over 15 min)	Model prompt drift or upstream Imagen 3 model behavior shift	1. Check Looker Evaluator Breakdown for lowest vector. 2. Inspect Model Armor logs for new false-positive blocks. 3. Temporarily route traffic to <code>high_fidelity.yaml</code> profile.
ALERT: VectorSearchDegraded	ScaNN Index sync delay or endpoint timeout	1. Confirm Circuit Breaker tripped to fallback Text-to-Image scratch generation mode. 2. Check Vertex AI Vector Search endpoint index deployment status in GCP Console. 3. Flushes pending embedding queue in Pub/Sub DLQ.
ALERT: BigQueryQuotaExceeded	Telemetry ingestion rate spike	1. In-memory buffer in <code>metrics_logger.py</code> automatically caches events in Redis. 2. Increase BigQuery streaming insert quota via GCP Console. 3. Flushes Redis backlog once quota recovers.
ALERT: RedisConnectionTimeout	Memorystore failover / network blip	1. Verify Circuit Breaker tripped to degraded mode (bypassing cache). 2. Confirm Cloud Run instances auto-reconnect via VPC Connector. 3. Run health check <code>curl -I https://api.imagesense.internal/readyz</code> .

Future Section for Potential Addendums

Deliverable Name	Description	Planned Phase
3D Garment Pose Alignment	Integration with 3D garment mesh estimators for extreme pose camera angle verification.	Phase 3 Addendum
Enterprise Terraform Module Registry	Centralized, versioned Terraform module registry with automated drift detection and policy-as-code enforcement.	Phase 3 Addendum
Multi-Region Active-Active DR	Cross-region Spanner / AlloyDB replication with global anycast routing across US and EU regions.	Phase 3 Addendum
Interactive Video Try-On (VTON)	Extension of the 5-vector evaluation flywheel to multi-frame video and animated fashion assets.	Phase 4 Roadmap